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Subjective Poverty and the Limits of Relative-Income Measurement: Evidence from Greece, 2003–2025

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ABSTRACT

Greece has reported the EU's highest subjective financial strain every year since 2011, while its official relative-income poverty rate (AROP) was only 4th-highest in 2025. The resulting 47.6-point gap is more than three times the EU's next largest. AROPE narrows it to 39.7 points but does not close it. Using Eurostat data for the EU27, 2003–2025, this paper first shows that AROP's moving national threshold understated Greece's crisis-era living-standard loss: a fixed, inflation-adjusted 2008 threshold approximately doubles measured poverty during the crisis. It then tests AROPE's multidimensional intuition using year-fixed-effects panel models and leave-one-country-out prediction. Housing and cash-flow strain narrow Greece's out-of-sample residual; replacing headline unemployment with long-term unemployment cuts it from 11.6 to 3.9 points. A fixed cumulative-exposure specification reaches −0.8, but the preferred validation repeats candidate selection and estimation inside every held-out-country fold. Under that full nested procedure Greece's residual is +2.70 points and its prediction-error rank is 19th of 27: eight countries are predicted less accurately. The robust result is therefore a family of duration and sustained-exposure measures, not one uniquely selected variable. The paradox also persists among salaried workers, while reporting-style tests show that a stable cultural premium cannot explain the crisis-era widening or its concentration in financial self-reports.

These associational country-level results imply that AROP should be reported with a fixed-standard threshold, AROPE or deprivation indicators, and duration-sensitive measures during prolonged macroeconomic collapse.
Keywords: subjective poverty; relative-income poverty; anchored poverty; Greece; Eurozone crisis; long-term unemployment; fiscal adjustment programs; institutional trust

CONTENTS

1. Introduction

By its own account, Greece is the most financially strained country in the European Union. In every wave of Eurostat's harmonized household survey since 2011, a larger share of Greek respondents has said they struggle to make ends meet than in any of the other 26 or 27 member states — in the most recent year, 67.2%, more than triple the EU average of 17.6%. Yet Greece's official income-poverty rate tells a materially less extreme story. AROP, the relative-income rate measuring the share of the population below 60% of national median income, is elevated but only the 4th-highest in the EU in the most recent year, and ranges between 1st and 10th over the whole 2007–2025 period. The gap between the two — 47.6 points in the most recent year — is more than three times the next-largest gap of any EU country under the same comparison (Bulgaria, 14.9 points), and is not a matter of degree: no other EU country combines a mid-table income-poverty rank with a subjective-poverty rank of 1st. This paper treats that gap as its central object, not as a puzzle to be dissolved by switching to a different official measure.

Greece's broader official poverty-or-social-exclusion rate (AROPE), which also counts severe material deprivation and very-low household work intensity, narrows the gap — to 39.7 points, still by far the largest of any EU country — but does not close it. AROPE is used in this paper as a secondary, motivating benchmark, not as a substitute for AROP: it is the EU statistical system's own acknowledgment that income poverty alone is too narrow a lens, and the published aggregate tables used here do not reveal its components' household-level overlap (EU-SILC microdata could support that analysis under controlled research access, but that lies outside this paper's aggregate-only pipeline), so this paper does not claim to reconstruct AROPE from any combination of the variables tested below. This caution has independent support in the literature critiquing AROPE's construction: Zieleńska and Wnuk (2024) argue the union design presents a technical, ostensibly neutral aggregation while concealing the normative choices and heterogeneity it combines, and Bárcena-Martín, Pérez-Moreno, and Rodríguez-Díaz (2020) show formally that countries with similar AROPE rates can conceal very different underlying multidimensional poverty profiles, precisely because a union indicator discards information about component severity and concurrence. What Part II of the paper does instead is test AROPE's underlying premise directly — that hardship is multidimensional — one dimension at a time, using AROP as the constant poverty covariate throughout.

This paper investigates how much of the AROP-based divergence reflects a real, measurable deterioration in living standards that a relative-income indicator is simply not built to capture, and how much of what remains is a genuine, still-open puzzle. It does so through two research questions:

> ***RQ1.*** *To what extent does Greece's exceptionally high subjective poverty reflect measurable deterioration in living standards that relative-income poverty does not capture?*
>
> ***RQ2.*** *How much of Greece's cross-country subjective-poverty gap can measurable financial hardship account for, and how much remains unexplained?*

The empirical core of the paper (§5–6) draws on a publicly reproducible Eurostat panel covering the EU27 and, where available, the EU aggregate, 2003–2025. Part I reconstructs a fixed, inflation-adjusted poverty threshold anchored to 2008 living standards and cross-validates it against independent household-microdata research. Part II fits a cross-country panel regression predicting subjective poverty from progressively richer sets of economic-condition variables, validated throughout with leave-one-country-out prediction rather than in-sample fit alone, and identifies first long-term unemployment, then cumulative excess unemployment accumulated since 2009, as the strongest and most stable explanatory markers of Greece's outlier status — associational markers whose robustness is established through out-of-sample, placebo, and nested-selection validation, not causally identified mechanisms.

To avoid ambiguity, the paper keeps AROP and AROPE in distinct, non-interchangeable roles throughout. AROP is the paper's primary poverty measure: it anchors the headline comparison, the moving-threshold mechanism (because its median-relative poverty line is the mechanism being tested), and the poverty covariate in every panel model. AROPE appears only as a secondary, official benchmark shown immediately alongside AROP to establish how much of the gap a broader definition alone can close, and as the conceptual motivation for Part II's multidimensional-hardship tests — never as the model covariate, since it partly bundles material deprivation and work-intensity components that Part II tests separately.

The argument that follows moves through nine stages, and it is worth naming them here so the reader can distinguish the paper's central claims from its supporting and contextual evidence throughout: the AROP puzzle itself (§5.1); a measurement artifact that explains part of it (§5.2–5.3, the moving poverty threshold); the AROPE bridge, which narrows but does not close the gap (§5.1); a decomposition of AROPE's underlying intuition, not a reconstruction of AROPE itself (§6, introduction); a residual that survives correcting for the threshold and testing multidimensional hardship (§6.1–6.3); the mechanisms that close nearly all of what remains — long-term unemployment (§6.5) and, more fully, cumulative excess unemployment (§6.6) — alongside material and social scarring that accompanied both (§6.7); supporting cross-country evidence that the puzzle persists even among salaried workers (§6.9, hourly work effort and reward); and one candidate explanation checked and ruled out (§6.10, income inequality); a dedicated robustness check (§6.11) then tests whether any of §5–6's results could instead reflect country-specific reporting style rather than material hardship. This ordering is separate from, and should not be conflated with, each finding's evidentiary status (core / descriptive / exploratory, §4.3): a finding can carry real weight in this argument's structure while resting on descriptive rather than model-tested evidence.

This paper's empirical contribution is Sections 5 and 6. Section 7 adds two further layers. The first is methodological: given how much of Greece's residual is ultimately explained by accumulated, rather than current-year, labor-market exposure, §7.1 argues that relative-income measures like AROP are safest interpreted alongside a fixed-standard threshold and a cumulative-exposure measure in prolonged macroeconomic collapses, not read alone. The second, in §7.3, is a brief, more speculative layer of interpretive context, not a second finding of equal standing: it places the §5–6 results alongside the documented history and design of Greece's three EU/IMF/ESM financial-assistance programs (2010, 2012, and 2015), asking whether that policy record is *consistent with* why long-term unemployment and unresolved scarring, rather than the crisis's depth alone, are what best explain Greece's persistent outlier status. The programs' design and the paper's own regression findings are never treated as a single tested causal chain; §7.3 states this explicitly and hedges accordingly throughout.

2. Institutional and macroeconomic background

Greece entered the euro area in 2001 following a decade of macroeconomic stabilization, and experienced a decade of debt-financed, consumption-driven growth thereafter, averaging 4.1% real GDP growth in 2000–2007 (Pagoulatos, 2018). This growth was underwritten by persistent primary fiscal deficits, a near-tripling of private-sector debt as a share of GDP, and a current-account deficit that reached 15% of GDP by 2008 — financed overwhelmingly by external capital inflows rather than domestic saving or foreign direct investment (Pagoulatos, 2018). The 2008 global financial crisis triggered a "sudden stop" of these inflows; a 2009 revision of the reported budget deficit from an official 6% to an eventual 15.1% of GDP eroded market confidence sharply, and by early 2010 Greece had lost market access entirely.

What followed was, on any conventional metric, one of the deepest peacetime economic contractions on record for an advanced economy: an eight-year-equivalent depression as steep as the United States' in the 1930s, but roughly twice as long (Pagoulatos, 2018). Real GDP fell by around a quarter from its 2008 peak; unemployment peaked at 27.8% in 2013 (Eurostat series for ages 15–74, used throughout this paper; Pagoulatos reports 27.5% on a different series vintage); three consecutive financial-assistance programs were required between May 2010 and August 2018, a distinction unique among euro-area crisis countries. A brief account of these programs' design and record is given in §7.3, as one contextual dimension for interpreting this paper's own empirical findings rather than as background scene-setting; readers uninterested in that connection can proceed directly to §3.

3. Related literature

3.1 The moving-threshold problem

The mechanism at the center of this paper's first research question — that a relative poverty line can understate a crisis because the line itself falls with the collapsing average it is drawn from — is well established, including for Greece specifically. Andriopoulou, Kanavitsa, and Tsakloglou (2019/2020) compute an anchored (fixed-threshold) poverty rate for Greece for 2007–2016 using household-level EU-SILC microdata, finding that a 2007-anchored measure shows substantially more deterioration than the contemporaneous relative AROP series; their income-year-2013 anchored rate of 48% of the population is used in §5.3 as an external validation benchmark for this paper's own, coarser reconstruction. Leventi and Matsaganis (2016), in an OECD working paper using the EUROMOD microsimulation model, independently document the same arithmetic: Greece's relative poverty rate moved only modestly across 2009–2014 (13.2% to 13.8%) despite a collapse in absolute living standards over the same years, precisely because the underlying median income fell in step. The OECD's own 2018 Economic Survey of Greece reaches the same conclusion through a different lens, noting that falling wages mechanically lowered the relative poverty threshold during the crisis, making some groups appear relatively better protected even as their real living standards deteriorated (OECD, 2018). Goedemé et al. (2022) propose a new EU-wide poverty indicator, the extended headcount ratio, designed specifically to address this same measurement problem, and use Greece as their lead illustrative case.

3.2 Subjective poverty as a research object

Subjective poverty measurement is an active, distinct literature with its own methodological conventions. Želinský, Mysíková, and Garner (2022) track subjective income-poverty rates across the EU using a "minimum income question" design, a different construct from the "ability to make ends meet" measure used here and by Eurostat's own headline indicator, but one that independently flags Greece and Bulgaria as extreme outliers. A related paper (Želinský, Ng, & Mysíková, 2020) derives an income threshold from make-ends-meet responses using the Youden index — a genuinely different construction from this paper's anchored-poverty measure, and a natural extension for future work (§8). Guagnano, Santarelli, and Santini (2016) model subjective poverty across Europe against household and community characteristics including social capital, a reminder that economic variables are not the only plausible predictors in this literature.

3.3 Cross-country gaps beyond economic fundamentals

Part II's central empirical puzzle — a country reporting persistently worse subjective outcomes than its measured economic conditions predict — has a structural parallel in the "happiness gap" literature comparing post-communist and Western European countries. Nikolova (2016) finds that gap is not fully explained by economic conditions, and that institutional quality (rule of law specifically) accounts for a further share of it. This is not evidence about Greece, and the outcome variable (life satisfaction) differs from subjective poverty, but it motivated this paper's own tests of household debt and welfare-transfer effectiveness as candidate residual explanations (§6.2), both of which returned null results. Separately, Claessens, Kyritsis, and

Atkinson (2023) demonstrate that cross-country regressions commonly treat national observations as more independent than they are, understating true uncertainty when neighbouring or culturally linked countries co-move; this paper's use of country-clustered standard errors partially, but not fully, addresses that concern (§8).

3.4 Housing tenure, scarring, and institutional trust

Filandri, Pasqua, and Tucci (2025) link housing tenure and rent regulation to subjective poverty among young European adults (18–34) using EU-SILC data across 24 countries, providing theoretical grounding for this paper's own housing-tenure findings (§6.7), though their result is age-specific and not shown here to generalize to the full population. Baldini, Gallo, and Torricelli (2020) find that a past income-scarcity spell continues to depress subjective make-ends-meet assessments up to two years later across Europe, even controlling for current income — the same scarring mechanism this paper documents for Greece over a far longer horizon (§6.7). Gourinchas, Philippon, and Vayanos (2016, 2017) characterize the Greek crisis as one of the most severe "trifecta" crises (sudden stop, sovereign debt, and lending boom-bust together) on record, with output collapse "significantly more severe and protracted" than any comparable post-1980 episode. On institutional trust specifically, Ervasti, Kouvo, and Venetoklis (2019) show the crisis damaged Greek trust in political and impartial institutions while leaving interpersonal trust largely intact, and Economou et al. (2014) link low institutional and interpersonal trust to worse mental-health outcomes during the crisis. On migration, Labrianidis and Vogiatzis (2013) argue skilled emigration and the debt crisis were mutually reinforcing rather than the former simply a consequence of the latter, and Pratsinakis (2022) cautions against reducing the post-2010 emigration wave to a single "brain drain" narrative, showing mixed motives among emigrants in an original survey.

3.5 Contribution of this paper

The contribution is the combination of three results: linking AROP's moving-threshold problem to subjective lived experience; testing AROPE's multidimensional intuition with out-of-sample country comparisons; and showing under full nested validation that duration-sensitive labor-market measures remove Greece's exceptional residual status. Long-term unemployment is the single substitution that most changes Greece's standing; the paradox persists among salaried workers; and reporting heterogeneity is tested directly rather than assumed away. These are associational findings developed through sequential model building, not pre-registered causal estimates. Adjustment-program history is interpretive context, not a separate empirical contribution.

4. Data and methods

All data underlying the paper's core empirical results (§5–6) are drawn live from the Eurostat SDMX-JSON dissemination API; no series in that core panel is hand-entered, and the full extraction and analysis pipeline is a numbered sequence of reproducible scripts (§10). This does not extend to §7: the institutional-trust figures cited there and in §6.7 are literature and survey context (OECD, 2026; underlying Eurobarometer time series), not part of the Eurostat pipeline or this paper's own models, and the policy-history material is drawn from two secondary sources read in full for this paper (Pagoulatos, 2018; European Stability Mechanism, 2020) — both cited in full in §12. The core panel covers subjective poverty (Eurostat series `ilc_mdes09` / `ilc_sbjp01`), AROP (`ilc_li02`), AROPE (`ilc_peps01n`), the poverty threshold in nominal terms (`ilc_li01`, deflated to real terms using cumulative Greek HICP, `prc_hicp_aind`) alongside real household disposable income already indexed to 2008 (`tepsr_wc310`), real GDP per capita (`sdg_08_10`), real wages (`nama_10_lp_ulc`, deflated by `prc_hicp_aind`), long-term and headline unemployment (`une_ltu_a` and the corresponding headline series), housing-cost overburden by tenure status (`ilc_lvho07c`), net migration of nationals (`migr_emi1ctz`, `migr_imm1ctz`), and, for §6.9, in-work subjective poverty, AROP, and AROPE by activity status (`ilc_sbjp03`, `ilc_iw01`, `ilc_peps02n`), actual weekly hours worked (`lfsa_ewhan2`), and compensation per hour worked (`nama_10_lp_ulc`, D1_SAL_HW). Coverage begins in 2003 for the earliest series and varies thereafter by indicator; exact year ranges are reported alongside each result. AROPE itself carries a methodology break at the 2020/2021 boundary (Eurostat revised the underlying material- and social-deprivation definitions that year); every AROPE figure and series in this paper splices the legacy series (`ilc_peps01`, through 2020) with the revised series (`ilc_peps01n`, from 2021) at that boundary rather than blending the two vintages, the same convention applied to the material-deprivation series it partly draws on.

4.1 Anchored-poverty reconstruction

Eurostat does not publish a pre-crisis-anchored poverty series for Greece. This paper approximates one by holding the 2008 nominal poverty threshold fixed and deflating it forward using Greek consumer-price inflation, then recomputing the poverty headcount against that fixed real threshold rather than against each year's own moving 60%-of-median line. The approximation is built from four published summary points rather than household microdata, and is validated in two ways (§5.3): against Eurostat's own 2019 anchored-poverty publication, and against Andriopoulou et al.'s (2019/2020) independent microdata-based reconstruction for 2007–2016. The method performs to within roughly one percentage point of the official benchmark on average, but is least certain in exactly the years (2012–2014) where it shows the largest effect — a limitation carried through into every result that relies on it.

4.2 Cross-country panel regression

Part II's central specification is an ordinary-least-squares panel regression of subjective poverty on a nested sequence of predictors, with year fixed effects and country-clustered standard errors (`cov_type='cluster'` , clustered by country), estimated across all EU member states with available data for each specification. Seven nested models are reported in Table 1 (§6.4), and an eighth specification — Model G, C-LTU plus cumulative excess unemployment — is introduced in §6.6 and completes the ladder in Table 2: Model A (headline unemployment, PPS-adjusted income, material deprivation, AROP); Model B (A + housing-cost overburden); Model C (B + arrears and inability to cover an unexpected expense); Model C-LTU (C, with long-term unemployment replacing headline unemployment); Model D (C + household debt-to-income and welfare-transfer effectiveness); Model E (C + financial expectations and household saving rate); and Model F (C + a GDP-scarring-stock variable). Every specification is evaluated two ways: in-sample fit, and leave-one-country-out (LOO) prediction, in which the model is refit on 26 countries and evaluated on the 27th — repeated for every country in turn, not only Greece, to establish whether Greece's out-of-sample gap is unusual relative to the distribution of all countries' own out-of-sample gaps under the same procedure.

Benjamini-Hochberg false-discovery-rate correction is applied within each family of related exploratory tests run together in this project: the level, year-over-year, and detrended correlation screen across the full 19-variable predictor set used in the companion technical report's own correlation table (17 of 19 survive on levels, 16 of 19 on year-over-year change, 17 of 19 detrended, corrected separately per column from full-precision p-values; reported in full in the companion technical report, not reproduced here), the year-over-year sensitivity-versus-residual test used in this paper (§6.7, 9 predictors, only real GDP survives), and a second-difference/acceleration check (5 variables, technical report only, only material deprivation survives). An earlier, differently-constructed 21-variable screen (contemporaneous level and one-year-lag correlations only, predating the year-over-year/detrended robustness columns) was also corrected at the time (16 of 21 survived) but has been superseded by the 19-variable screen above for every claim in this paper and the technical report; it is retained only as an earlier exploratory record. No single correction is applied across the whole project, since these families answer different questions and were not run as one combined test; the panel regressions' own coefficients (e.g. each model's added-variable p-value in Table 1) are each a single test of one stated hypothesis within its own specification, and are reported as such rather than folded into any FDR family. They are not, however, pre-registered: this paper's specifications were developed sequentially across a series of checkpoints, so an added-variable p-value should be read as one test within a nested sequence the analyst had already seen, not as a pre-specified confirmatory test. That is precisely why the results in §6.5–6.6 rest on out-of-sample, placebo, and nested cross-validation rather than on those coefficients alone (§4.3).

4.3 Evidentiary hedging convention

Throughout §5–6, findings are explicitly labeled by evidentiary status rather than left to be inferred from tone, across four tiers that should not be conflated: *pre-planned confirmatory tests* (one variable added to a fixed specification to test one stated hypothesis — the paper's core findings where robust to the leave-one-country-out check); *exploratory screening* (multi-candidate sweeps, FDR-corrected within their family — noting that FDR correction controls false discoveries inside a family but does not convert sequentially developed specifications into pre-specified tests); *post-selection robustness* (validation of an already-selected variable — leave-one-out stability, placebo inference, and the nested re-screening of §6.6 — which establishes stability but not pre-specification); and *descriptive corroboration* (real, verified patterns reported without a model test, including variables conceptually too close to the outcome to be treated as independent predictors, and patterns surviving only a weaker standard of evidence, flagged as exploratory leads). The later scorecard rows (C-LTU, Model G) accordingly rest their claim on out-of-sample, placebo, and nested-selection validation rather than on in-sample p-values alone. No result in this paper is described as causally identified; the panel regressions establish association conditional on observed covariates and year fixed effects, not a causal estimate, and this is stated explicitly wherever a finding might otherwise be read more strongly than the design supports.

This evidentiary labeling is deliberately kept separate from the nine-stage argument structure set out in §1. The two answer different questions: evidentiary status asks how much statistical or methodological weight a given result can bear; argument stage asks why that result is being presented at that point and what work it does for the paper's overall claim. A finding can be central to the argument (e.g., material scarring in §6.7) while itself resting on descriptive rather than model-tested evidence — that combination is common in this paper and is not a weakness to be smoothed over: it means the finding is doing real narrative work without claiming statistical strength it doesn't have. Where a result's role in the argument might otherwise be ambiguous, its stage is named directly in the surrounding prose (e.g., "material-scarring context," "ruled out") alongside, not instead of, its evidentiary label.

4.4 Limitations of the data

All results rely on Eurostat's published, country-level aggregate series rather than household microdata. This means the analysis cannot observe within-country heterogeneity — who, specifically, is driving Greece's outlier status, or whether it is concentrated in particular household types — and cannot rule out that some of the residual gap reflects compositional factors (household structure, informal income, savings buffers) invisible to country-year aggregates. Where a candidate mechanism specifically requires microdata to test properly (for example, the Youden-index subjective-poverty-line construction of Želinský et al., 2020), this paper flags it as a natural extension rather than attempting an aggregate-data approximation that would not meaningfully test the same claim.

5. Results I — Does subjective poverty reflect real hardship?

5.1 The paradox, documented

Across all 27 EU countries plus the EU aggregate, Greece's gap between AROP (the EU's standard relative-income poverty rate) and subjective poverty is in a different league from every other country's, in either direction: 47.6 percentage points, against a next-largest gap (Bulgaria) of 14.9 points — more than three times the runner-up. Most EU countries sit far closer to agreement between the two measures, in either direction; Greece is a singular exception, not one extreme of a continuous spread. The divergence also predates the debt crisis: in 2003, 48% of Greek households already reported difficulty making ends meet, more than double the contemporaneous AROP rate of 20.7%.

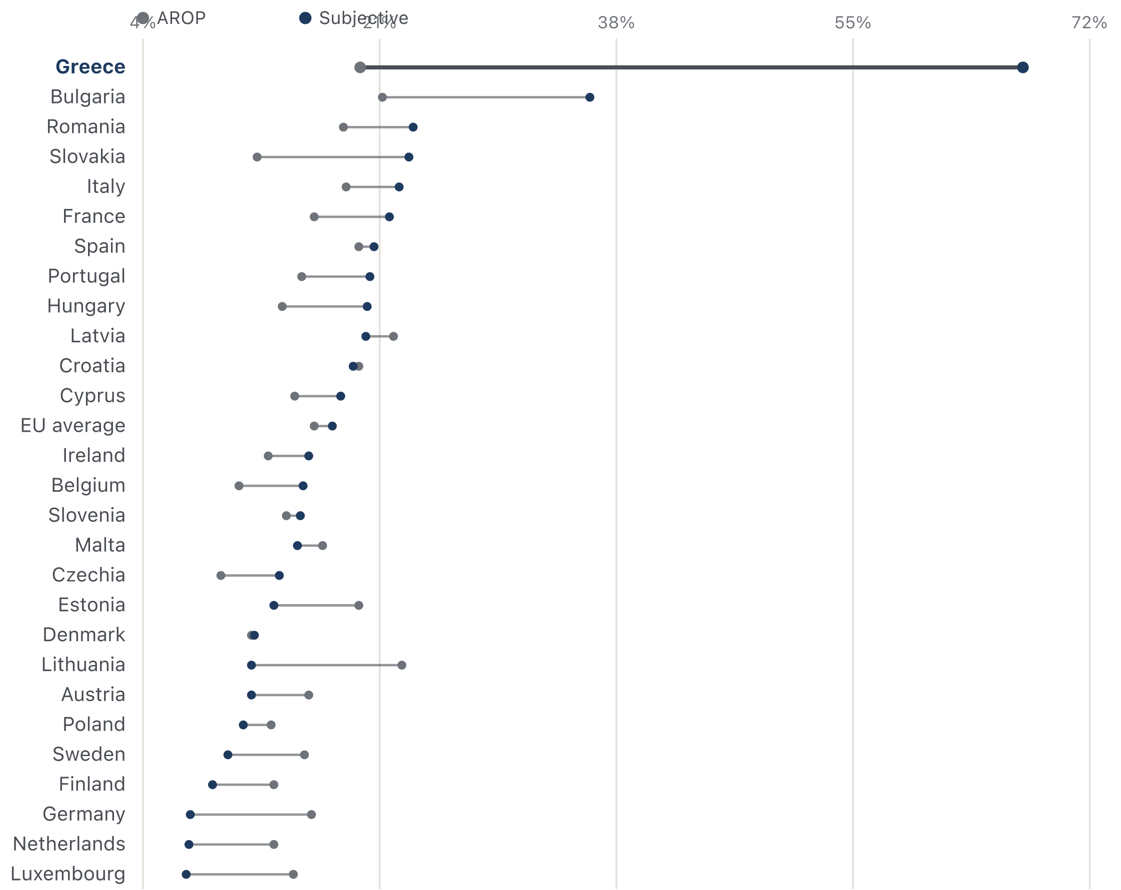

Figure 1. AROP and subjective poverty by EU country, most recent year available. Sorted by subjective poverty. Source: Eurostat `ilc_li02`, `ilc_sbjp01`.

Greece's broader official measure, AROPE, is worth reporting alongside AROP here because it is the EU statistical system's own acknowledgment that income poverty alone is too narrow. It narrows Greece's gap from 47.6 points to 39.7 — still the largest gap of any EU country by a wide margin, next-highest Bulgaria's 7.1 points under the AROPE comparison — but does not close it. This paper does not treat AROPE as interchangeable with AROP: because the published aggregate tables used here do not reveal how AROPE's three components (income poverty, severe material deprivation, very-low work intensity) overlap within a single household, no combination of the variables tested in §6 can be presented as a reconstruction of AROPE itself. What Part II tests instead is AROPE's underlying premise — that hardship is multidimensional — one dimension at a time, with AROP retained throughout as the model's poverty covariate.

5.1.1 Breadth of relative disadvantage (descriptive)

Greece's disadvantage is not confined to the two indicators in Figure 1. Define, for each country-year, the share of indicators on which that country falls in the worst quintile of the EU distribution. The indicator set comprises 25 series with an unambiguous worse-direction, declared ex ante, spanning the labour market, output, wages, prices relative to wages, housing conditions, demography and household expectations. The outcome variable and all seven Model C-LTU covariates are excluded by construction, so the measure cannot reproduce the quantity the paper seeks to explain.

For Greece this share averages 21% over 2005–2010 and 58% from 2012 onward, reaching 68% in 2024 — first of 27 member states, ahead of Bulgaria at 50%. The discontinuity is located at 2011 and does not reverse over the subsequent fourteen years. The ordering is insensitive to the quintile cut-off: Greece ranks first of 27 at the 10th, 20th, 25th and 33rd percentile definitions of "worst", and the pre- to post-crisis increase holds at each (6% → 50% at the strictest, 41% → 67% at the loosest).

Evidence tier. Descriptive corroboration (§4.3), not an identified channel. The measure was operationalised as a regressor and tested within a pre-registered family of six persistence variables; it does not survive multiple-testing correction (p=0.083 raw, 0.287 after Benjamini–Hochberg) and does not improve Greece's out-of-sample residual. It is reported because it characterises the condition compactly, not because it explains it, and it is excluded from every specification in Section 6. Construction and the null result are in `scripts/50_persistence_share.py`; the full country-by-country series is in the statistical appendix, together with a per-indicator ladder giving Greece's position in the EU distribution on each of the 25 at both ends of its observed span.

5.2 The moving-threshold mechanism

Greece's real, inflation-adjusted poverty threshold fell from an index value of 100 in 2008 to 65 in 2016 — a 35-point real decline that tracks the collapse in real household income (which fell to 68 over the same span) almost exactly. Because the poverty line is defined relative to the (collapsing) national median rather than to a fixed living standard, the measured AROP rate barely moved over this period, rising only from 20.1% to 21.2%, even as the population beneath the line grew markedly poorer in real terms.

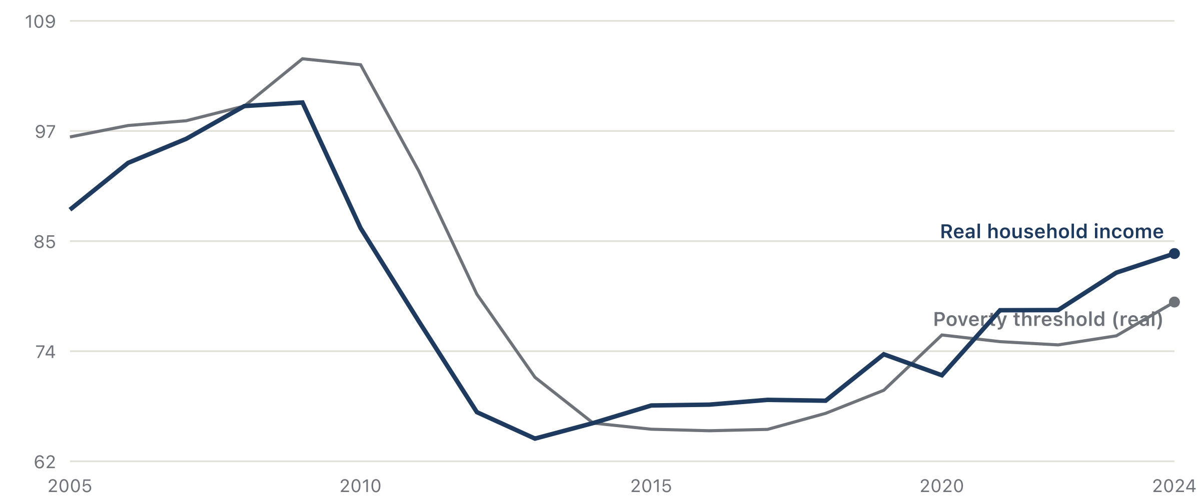

Figure 2. Greece's real (inflation-adjusted) AROP poverty threshold and real household income, both indexed to 2008=100. Source: Eurostat `ilc_li01`, deflated with Greek HICP (`prc_hicp_aind`); real household income from `tepsr_wc310`.

5.3 Anchored-poverty reconstruction and external validation

Recomputing the poverty headcount against a fixed, 2008-anchored real threshold rather than the moving AROP line shows measured poverty roughly doubling during the debt crisis, tracking subjective reports of hardship far more closely than AROP does over the same period. This paper's own reconstruction (survey-year 2014: 40.6% of the population below the anchored threshold) is conservative relative to Andriopoulou et al.'s (2019/2020) independent microdata-based estimate for the closest reported year (income year 2013: 48%), a 7–8 point gap attributable to differences in data source (aggregate summary points versus full household microdata), anchor year (2008 versus 2007), and price-index construction — and potentially to the survey-year/income-year distinction itself, which this paper has not independently verified against Andriopoulou et al.'s exact reference-period convention. None of these can be cleanly isolated as the dominant cause. The direction of the gap — the independent, more granular study finds an even larger crisis-era effect than this paper's coarser approximation — is itself informative: if anything, this paper's Part I findings understate the case for a real, measurable deterioration that AROP fails to register.

5.4 Does the relationship hold over time and across countries?

Within Greece, subjective poverty tracks the anchored-poverty reconstruction closely across levels, first differences, and a detrended series (full correlation table in the companion technical report, §10). A within-country, two-way fixed-effects panel across the EU27 shows the same directional relationship generalizes beyond Greece, though with substantially more noise than the Greece-specific series alone — consistent with Greece representing an unusually clean and unusually severe instance of a mechanism that operates, more weakly, EU-wide.

> **ANSWER TO RQ1**
>
> A substantial, real part of Greece's subjective-poverty exceptionalism is arithmetic: because the standard EU poverty line resets each year against a collapsing median, it systematically understated the depth of Greece's crisis-era income losses. Holding that line fixed at a pre-crisis standard reveals a doubling of measured poverty that AROP does not show, corroborated by an independent, more granular microdata study finding an even larger effect. This is not the whole explanation: even after this correction, Greece's cross-country subjective-poverty gap does not fully close, motivating Part II.

6. Results II — Why does Greece remain an outlier?

6.1 Baseline cross-country residual

A baseline model (Model A: headline unemployment, PPS-adjusted income, material deprivation, AROP) predicting subjective poverty across all EU countries, 2015–2024, leaves Greece with the largest average unexplained residual of any EU country — ahead of Cyprus and Croatia, the next-highest.

6.2 Housing and cash-flow strain

Adding housing-cost overburden (Model B) and then arrears and inability to cover an unexpected expense (Model C) reduces Greece's in-sample average residual from 15.1 points to 8.8 and then to 2.2 — near zero for most of 2015–2019, though a roughly 6-point gap re-emerges by 2024, concentrated in the 2020s. Household debt-to-income and the poverty-reducing effect of social transfers were tested as further additions and returned coefficients statistically indistinguishable from zero once housing and cash-flow strain were already in the model — a narrower finding than ruling either factor out entirely, since their effects could be absorbed by the variables already present.

6.3 Out-of-sample validation

Because every model above includes Greece in its own training data, its own pattern could be inflating the apparent fit. Refitting each nested specification on the other 26 EU countries and evaluating it on Greece alone — and repeating this leave-one-out procedure for every country, not only Greece, to establish a baseline for what a "normal" out-of-sample shift looks like — tells a different, less monotonic story. Going from the basic comparison to "+ housing" does not shrink Greece's out-of-sample gap the way it does in-sample; it grows, from 25.6 to 35.5 points. Only once cash-flow strain (arrears, unexpected-expense capacity) is added does the gap fall sharply, to 11.6 — still the largest of 27, but only narrowly ahead of Luxembourg (10.7) rather than clear of the field. The gap between the smooth in-sample decline and this more volatile out-of-sample pattern is itself informative: part of why the in-sample fit looks so clean is that those models are permitted to see Greece's own data while being estimated.

6.4 The full scorecard

Table 1. Nested model comparison, Greece's average residual, 2015–2024.

MODEL	R^2	IN-SAMPLE GAP	OUT-OF-SAMPLE GAP	OOS RANK
A: unemployment, income, deprivation, AROP	0.76	15.1	25.6	1st / 27
B: A + housing cost overburden	0.78	8.8	35.5	1st / 27
C: B + arrears, unexpected-expense capacity	0.89	2.2	11.6	1st / 27
C-LTU: C, long-term unemployment replaces headline	**0.91**	**1.3**	**3.9**	**6th / 27**
D: C + debt-to-income, transfer effect	0.89	2.2	13.0	1st / 27
E: C + financial expectations, saving rate	0.90	1.3	3.6	10th / 27
F: C + scarring stock (% below own GDP peak)	0.90	1.3	7.1	4th / 27

All specifications: year fixed effects, country-clustered standard errors. Out-of-sample = each country's own leave-one-country-out average residual.

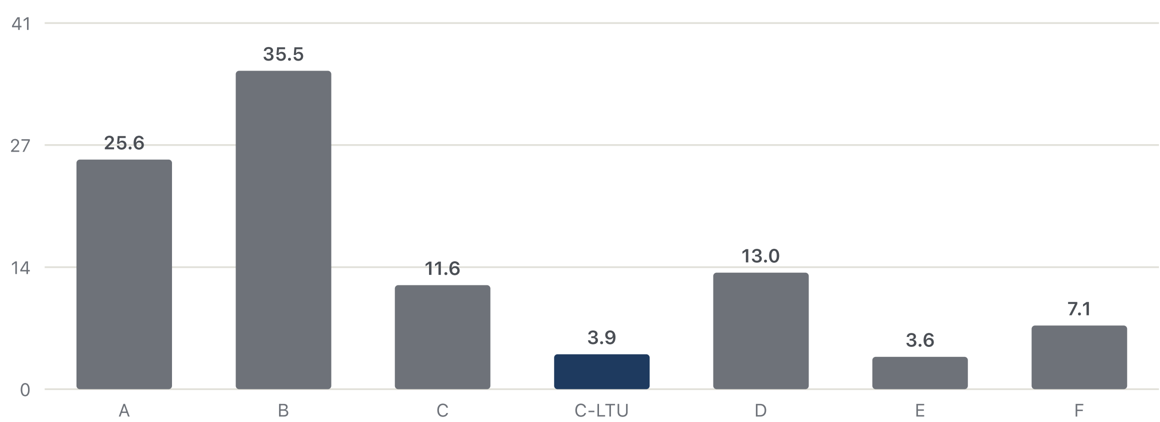

Figure 3. Greece's out-of-sample (leave-one-country-out) average residual, points, by model. C-LTU (highlighted) is the preferred specification. Corresponds to the out-of-sample column of Table 1.

6.5 Long-term unemployment as the sharpest current-conditions marker

The single most consequential result in this paper is not an added variable but a substitution: replacing headline unemployment with long-term unemployment (12 months or more out of work) in Model C. Long-term unemployment tracks Greek subjective poverty more tightly than any other variable tested, at every time scale — level ($r=0.93$, $p<0.0001$), first-difference ($r=0.92$, $p<0.0001$), and detrended ($r=0.86$, $p<0.0001$), all robust to false-discovery-rate correction. Substituted into Model C, it lifts the fit from $R^2=0.89$ to 0.91 and cuts Greece's out-of-sample gap from 11.6 to 3.9 points, moving Greece from the EU's largest unexplained outlier to 6th of 27. Its coefficient is stable and significant at $p<0.001$ across all 27 leave-one-country-out refits, including the refit excluding Greece itself, indicating the result is not an artifact of Greece's own data. Stability under refitting is distinct from precision under inference: conditional on accumulated unemployment and the other objective predictors, LTU is estimated imprecisely under wild-cluster inference ($p=0.073$), while accumulated exposure remained supported across all bootstrap variants ($p_{MC}=0.0005$ in the primary restricted specification, the 1/2,000 resolution floor; worst robustness result $p_{MC}=0.0070$, 6 exceedances of 999 draws, across Rademacher, Webb and Mammen weights and three seeds). Two p-values differing does not establish that the coefficients differ; the defensible statement is that the evidence is strongest for accumulated labour-market history. A robustness check adding long-term unemployment alongside (rather than in place of) headline unemployment achieves a marginally better fit but destabilizes headline unemployment's coefficient (sign flip, loss of significance, variance inflation factor ≈8–9), a textbook symptom of collinearity that motivates treating this as a replacement rather than an addition. Greece's own long-term unemployment rate peaked at 17.5% of the labor force in 2014 and has since fallen to 5.4% in 2024 — still the EU's highest by a wide margin (next: Spain, 3.8%).

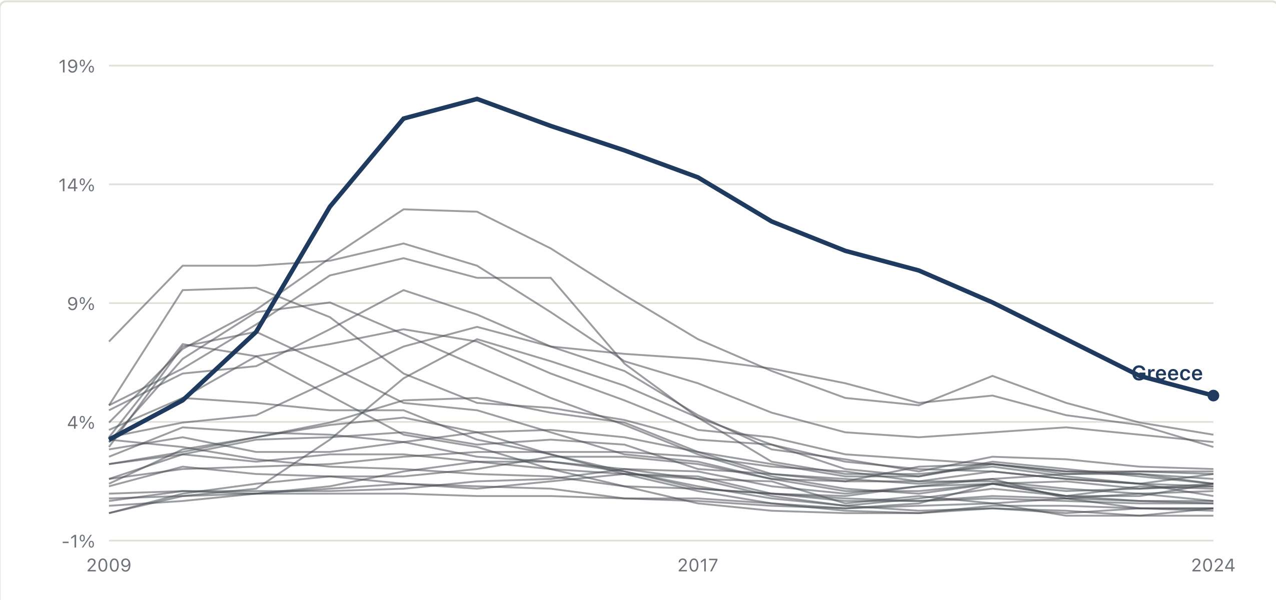

Figure 4. Long-term unemployment rate (12 months or more, % of active population), all EU countries, 2009–2024. Source: Eurostat `une_ltu_a`.

6.6 Sustained labor-market exposure removes Greece's exceptional residual

Current unemployment and even long-term unemployment remain snapshots. The paper therefore tests whether hardship is better associated with sustained exposure: each country's annual unemployment above its own 2009 baseline, floored at zero and accumulated through 2024. The fixed Model G specification reduces Greece's out-of-sample residual from 3.9 to −0.8 points. Because that variable was selected after screening related duration measures, the primary result is the fully nested procedure: candidate selection and estimation are repeated without the held-out country. Greece's nested residual is +2.70 points, ranking 19th of 27, with eight countries predicted less accurately. Greece therefore ceases to be an exceptional outlier without requiring the claim that one uniquely chosen variable explains the gap.

Table 2. Gap-closing ladder: from the raw AROP gap to the fixed final model, Greece — all rows on one common 2015–2024 window.

STEP	LAYER	GREECE'S GAP	POINTS CLOSED
1. Raw AROP gap	—	52.6	—
2. Raw AROPE gap	broader official measure	41.5	11.1
3. Basic model (A)	income, deprivation, unemployment, AROP	25.6	27.0
4. Model C	+ housing and cash-flow strain	11.6	41.0
5. Model C-LTU	duration, not just rate	3.9	48.7
6. Fixed Model G	**+ accumulated exposure**	**−0.8**	**53.4**

Steps 1–2 are raw gaps; steps 3–6 are out-of-sample model residuals, so this is a common-window narrative sequence rather than an additive decomposition. The fully nested result (+2.70; rank 19/27) is the primary validation result and is not another additive row.

The result survives correction within the 18-candidate screening family and is reproduced by a roughly ten-year rolling exposure window and by decayed alternatives; three- to five-year windows do not reproduce it. The defensible interpretation is therefore sustained exposure over approximately a decade, not literal permanent accumulation from one uniquely correct baseline. Wage-duration is the closest competing construction and is selected when Greece is excluded from the screening fold, which is why the paper treats the duration/exposure family as identified more clearly than any single member. Appendix A reports the complete screening, false-discovery-rate, replacement, nested-selection, and memory-horizon batteries.

This remains a country-level association, not evidence that the same households were continuously unemployed. It is consistent with individual-level scarring research but cannot substitute for household employment histories. The conclusion is correspondingly bounded: once sustained labor-market exposure enters the model, the hypothesis of a uniquely Greek unexplained pessimism premium is no longer supported.

6.7 Structural corroboration

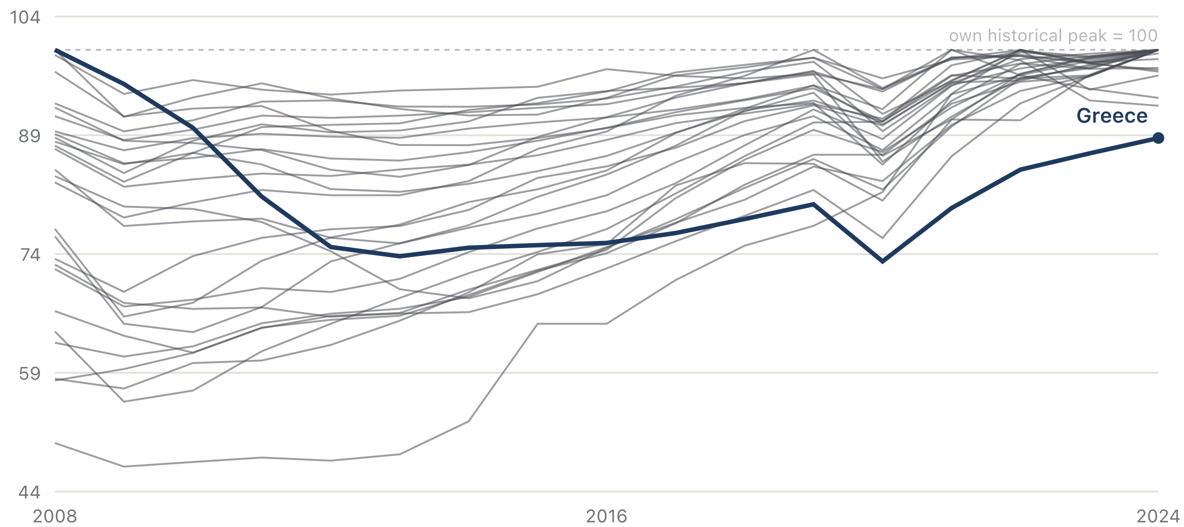

Figure 5. Real GDP per capita, indexed to each country's own historical peak (2008–2024). Greece has the deepest and longest unresolved pre-crisis shortfall; Finland remains slightly below 2008 and Luxembourg is below a recent peak. Source: Eurostat `sdg_08_10`.

Material scarring. Greece's real GDP per capita remains 11.2% below 2008, the EU's deepest and longest unresolved pre-crisis shortfall. Finland also remains 2.3% below 2008; Luxembourg is below a recent 2021 peak. The scarring-stock predictor is significant alone ($p=0.045$) but not beside long-term unemployment ($p=0.61$).

Real wages remain 31.8% below 2008, the EU's largest shortfall. Greece also records the highest wage-adjusted price pressure in every category tested despite only the 18th-highest raw price level, and 25.7% of mortgage-free owners face housing-cost overburden, the EU's highest rate. These findings document material context; they are not additional preferred-model mechanisms.

Social and demographic scars. Net emigration of Greek nationals totaled 290,281 people (2.6% of the 2008 population) between 2008 and 2024, peaking at 44,502 in 2012, with a net inflow only appearing in 2023–2024. Institutional trust in national government, tracked via Eurobarometer, fell from an already-middling pre-2008 average of 44% to a historic low of 7% in 2012, recovering only to 37% at its subsequent peaks (2015, 2020) before declining again to 24% by 2026 despite genuinely improved recent macroeconomic performance — a disconnect OECD's (2026) own account treats as unresolved. None of these trust or migration series are entered directly into the regression; they are reported as descriptive, independently corroborated context for the scale and duration of the underlying scarring, distinct in argument stage from the material-scarring findings above though comparable in evidentiary status (§4.3).

6.8 A generational reversal behind the aggregate AROPE bridge

Eurostat publishes AROPE, AROP, and severe material deprivation broken down by age, and this breakdown reveals a more specific story than a modest national uptick. This is a within-Greece distributional finding, not a further candidate explanation for the cross-country residual above; no age or household-composition variable is entered into any regression in this paper. It is reported here because it sharpens the AROPE bridge introduced in §5.1: the national AROPE figure that narrows the AROP gap is itself an average across generations whose experiences diverged sharply after 2020.

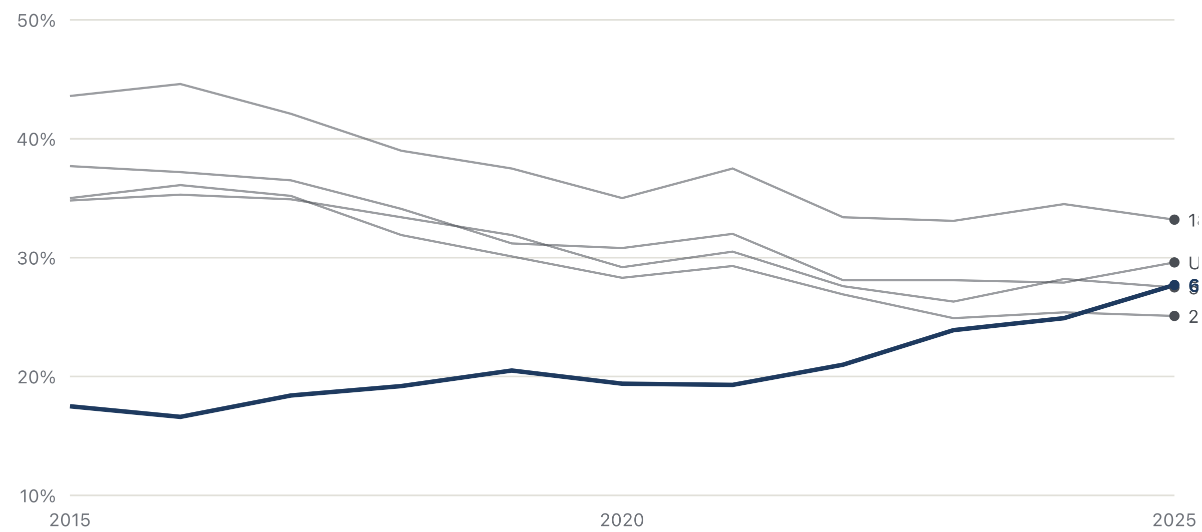

Figure 6. AROPE by age group, Greece, 2015–2025. Every working-age group declines; 65+ reverses course from 2020 onward. Source: Eurostat `ilc_peps01n`, age breakdown.

Every working-age group's AROPE fell over 2015–2025: under-18 from 37.7% to 29.6% (with a recent reversal, 27.9% in 2024 to 29.6% in 2025), 18–24 from 43.6% to 33.2%, 25–49 from 35.0% to 25.1%, 50–64 from 34.8% to 27.5%. People 65+ moved the opposite direction, from 17.5% (2015, the lowest of any age group) to 27.7% (2025, higher than every working-age group except 18–24), with the sharpest single-year increase between 2024 and 2025 (24.9% to 27.7%).

A shift-share decomposition of Greece's 2024-to-2025 national AROPE change (+0.6pp), using each age group's own population share (derived from Eurostat's AROPE persons-in-thousands series alongside the rate series, not assumed) separates a within-group rate-change effect from a between-group demographic-composition effect. The composition effect is +0.02pp — the population is not meaningfully aging into a higher-risk bracket year over year. The rate effect is +0.60pp, and decomposes further by age group: under-18 +0.28pp, 18–24 −0.09pp, 25–49 −0.09pp, 50–64 −0.15pp, 65+ **+0.65pp**. The 65+ group's own rate increase alone exceeds the entire net national change, meaning it is offsetting genuine improvement across every working-age group rather than simply adding to a shared deterioration.

Table 3. People 65+, Greece: AROP, severe material deprivation, and AROPE, selected years.

YEAR	AROP, 65+	DEPRIVATION, 65+	AROPE, 65+	AROPE, WHOLE POPULATION
2015	13.7%	7.1%	17.5%	32.4%
2020	13.0%	10.6%	19.4%	27.4%
2024	18.8%	12.8%	24.9%	26.9%
2025	**20.9%**	**14.1%**	**27.7%**	**27.5%**

Very-low-work-intensity, AROPE's third component, has no published breakdown for this age group at all — structurally undefined for a population largely outside the labor force, confirmed directly against Eurostat's own dimension codes rather than assumed. The 65+ increase can only be coming from AROP, deprivation, or their overlap.

Within the 65+ population, the deterioration concentrates specifically among women and people living alone, checked against two genuine Eurostat cross-tabs (age × sex, and household composition), not inferred by combining separate marginal tables. Women 65+: AROPE rose from 18.7% (2015) to 30.9% (2025), +12.2pp; men 65+: 15.9% to 23.6%, +7.7pp; the within-Greece gender gap widened from 2.8pp to 7.3pp over the decade. At EU27 level over the same span, both rose only modestly by comparison — women 65+ AROPE from 20.6% to 21.2% (+0.6pp) and men's from 14.7% to 15.8% (+1.1pp) — Greece's elderly women are diverging sharply from the EU pattern, not simply worse off in the same direction. By household composition (Eurostat's own `A1_GE65`, one adult 65+, and `A2_GE1_GE65`, two adults with at least one 65+): single-person 65+ households rose from 23.0% (2015) to 38.7% (2025), +15.7pp — the largest movement of any group or subgroup in this checkpoint; 65+ couples rose from 18.2% to 26.2%, +8.0pp. Greek single-elderly households started 2015 below the EU27 average (23.0% vs. 26.9%) and are now above it (38.7% vs. 30.0%), a reversal rather than a persistently worse position drifting further behind in parallel with the EU.

Asymmetric coverage windows, by design. Age-specific AROP is reported from 2003 because `ilc_li02` carries no legacy/revised definitional break at all. Age-specific AROPE and severe material deprivation are reported from 2015 only, because Eurostat's revised age-breakdown series (`ilc_peps01n`, `ilc_mdsd11`) begins there; the legacy series is not spliced in to extend the window, because a direct overlap check (legacy `ilc_peps01` vs. revised, 2015–2020) finds age-specific gaps too large to trust — up to 7.3 percentage points for 18–24 and 5.4 points for 65+, well beyond the roughly 1–3-point gap this paper already tolerates when splicing the whole-population AROPE series (§4). For context, not as a revision to this section's central comparison: Greek 65+ AROP was 29.4% in 2003, fell to a low of 11.6% in 2018, then rose for seven consecutive years to 20.9% in 2025 — still below its 2003 level. The current 65+ deterioration is a substantial partial reversal of an earlier, larger improvement, not a movement into unprecedented territory; that longer arc is background, and the 2015–2025 window remains this section's load-bearing comparison, since it is the only span over which AROP, AROPE, and deprivation can all be read on one consistent definition. Full overlap-check table: `age_breakdown_legacy_revised_overlap_check.csv`.

What was checked and found not to be a genuine Eurostat statistic, reported as absent rather than approximated. Three checks were run and did not yield a usable result: (1) sex within single-elderly households — the household-composition tables (`ilc_peps03n` , `ilc_li03`) carry no sex dimension, and the single-female/single-male categories in the same tables are not age-restricted, so combining them with the 65+ household category would construct a cross-tab Eurostat does not publish; (2) age-by-tenure as an outcome rate — the housing-cost-overburden-by-tenure table has no age dimension, and the one table that crosses household composition with tenure (`ilc_lvho02`) is a population-distribution table, not an overburden-rate table, so it cannot support a claim about overburden rates for a specific age/tenure combination; (3) formal sampling uncertainty — the Eurostat dissemination API's SDMX-JSON `status` field, where reliability or break flags would appear, was empty for every query in this section, checked directly rather than assumed absent. The qualitative caution in its place: Greece's single-65+ population, while real, is a modest 273,000 people as of 2025, so single-year movements for this and smaller subgroups carry more ordinary EU-SILC sampling variance than the national headline rate. The decade-long trend (2015 to 2025) is this section's load-bearing claim; the sharp 2024-to-2025 movement is corroborating, not independently definitive on its own. Full derivation: `scripts/39_age_breakdown_arope.py`; output tables `age_breakdown_arope.csv`, `age_breakdown_arop.csv`, `age_breakdown_deprivation.csv`, `age_breakdown_low_work_intensity.csv`, `age_breakdown_eu_arope_rank_2025.csv`, `age_breakdown_shiftshare_decomposition.csv`, `age_breakdown_65plus_by_sex.csv`, `age_breakdown_household_arope.csv`, `age_breakdown_household_arop.csv`, `age_breakdown_legacy_revised_overlap_check.csv`.

6.9 Hourly work effort and reward: the puzzle persists among salaried workers

Sections 6.1–6.8 locate this paper's residual within labor-market *duration* — long-term and cumulative excess unemployment. A separate question is whether the paradox survives entirely apart from unemployment status: does having a job resolve it? For salaried workers specifically, restricted to Eurostat's in-work subjective-poverty, AROP, and AROPE series (`ilc_sbjp03`, `ilc_iw01`, `ilc_peps02n`, wstatus=SAL), it does not.

Table 4. Salaried workers only, 2025: official poverty measures vs. subjective poverty, Greece vs. EU27.

	GREECE	EU27	GREECE'S EU RANK
AROP	5.7%	6.5%	17th of 27
AROPE	12.5%	9.2%	3rd of 27
Subjective poverty	59.5%	13.6%	1st of 27

Greek salaried workers rank an unremarkable 17th of 27 on AROP — squarely mid-distribution — while ranking 1st, by the widest margin in the EU, on subjective poverty. This reproduces §5.1's central paradox within a subpopulation from which unemployment has been removed entirely, addressing directly the possible objection that the paper's residual is primarily an unemployment story that §6.5–6.6 merely re-describes at the aggregate level.

One candidate explanation specific to the employed population: an imbalance between hours worked and hourly reward. Greece records the EU's longest actual weekly hours among all employed people (39.8 vs. an EU27 mean of 36.0), the longest full-time weekly hours (41.1 vs. 38.8), the 4th-highest national-accounts annual hours per employee (1,900 vs. 1,547), and, on the reward side, the lowest purchasing-power-adjusted hourly compensation of any EU member state (14.2 vs. 29.6). A composite index — relative weekly hours divided by relative PPS hourly compensation, EU27=100 — places Greece at 230.5, the highest in the EU; the pattern holds restricting hours to salaried employees alone (rank 7th of 27 on hours), ruling out Greece's comparatively large self-employed population as the sole explanation. Real hourly compensation, indexed to each country's own 2008 level and HICP-deflated, remains 27.8% below its pre-crisis level as of the latest year.

Added to Model C-LTU, this work-effort-squeeze index is significant after Benjamini-Hochberg correction across the pre-specified nine-candidate family this section screened (coefficient 0.078, raw p=0.00048, FDR-adjusted p=0.0030) and moves Greece's out-of-sample residual from +3.9 (rank 6 of 27) to −3.5 (rank 21 of 27); a matched national-accounts employee-hours construction also survives correction (FDR-adjusted p=0.0063). On the same simple bridge specification used in §5.1 (subjective poverty ~ AROP + candidate + year fixed effects), the index alone closes 13.8 points of Greece's 52.6-point average (2015–2024) raw AROP gap.

Two robustness checks distinguish this from §6.5–6.6's dynamic mechanisms rather than treating it as equivalent evidence. Adding country fixed effects to the same specification renders the coefficient non-significant (p=0.34); re-estimating on year-over-year first differences does the same (p=0.73); and within Greece's own 2010–2024 time series, the index and subjective poverty are essentially uncorrelated (r=−0.03, p=0.92). Long-term and cumulative excess unemployment, by contrast, track Greece's own within-country movement directly (§6.5–6.6). This paper therefore does not add the work-effort squeeze to the §6.4 scorecard alongside those variables: it is a genuine, FDR-corrected, cross-country structural association — evidence about a stable feature of where Greece sits relative to the rest of the EU — not a dynamic mechanism explaining year-to-year change, and the two kinds of evidence are kept explicitly distinct rather than pooled into a single claim of explanatory strength. A secondary exploratory result, not part of the pre-specified FDR family and reported with appropriate caution given a maximum variance-inflation factor of 9.05 among its regressors: an interaction term between standardized hours and standardized low hourly pay is independently significant (coefficient 1.52, p=0.024), consistent with, though not independent confirmation of, the same effort-reward account.

Full derivation, the salaried/self-employed/all-employed decomposition, the nine-candidate FDR-corrected battery, and both within-country robustness checks are reproducible from `scripts/43_work_effort_squeeze.py`, logged in the companion technical report's `publication_strategy.md`.

6.10 Candidate explanations ruled out or not supported

Income inequality (S80/S20 and Gini) was tested directly as a candidate residual explanation and found not to contribute: Greek inequality is elevated but not exceptional (6th-highest of 27 EU countries), and adding it to Model C-LTU explains none of the remaining residual. A cumulative shortfall in the AROP poverty threshold itself — despite being the most narratively anticipated candidate given the moving-threshold finding of §5.2 — was tested directly (§6.6) and returned a null result ($p=0.291$).

ANSWER TO RQ2

Nearly all of Greece's cross-country subjective-poverty gap is accounted for once sustained labor-market exposure, not just current conditions, is included — but reaching that result requires several layers, each tested and reported separately rather than assumed. In-sample, housing and cash-flow variables shrink the basic gap from 15.1 to 2.2 points. Out-of-sample — the stricter, more honest test — the same variables narrow the raw 47.6-point AROP gap to 11.6 points. Replacing headline with long-term unemployment narrows it further, to 3.9. Adding cumulative excess unemployment since 2009 — each country's accumulated exposure above its own baseline, not merely its current rate — closes what remains: Greece's out-of-sample residual moves to −0.8, meaning the final model slightly overpredicts Greece's reported hardship. This coefficient is robust across all 27 leave-one-country-out refits and holds excluding Greece from the fit entirely; a closely related duration measure, years spent below Greece's own 2008 real-wage level, is close enough behind it (and edges ahead once Greece is excluded from candidate selection itself, not merely from coefficient re-estimation) that the two are best read jointly, not as one clearly primary finding (§6.6). A further check rules out that this is merely a proxy for elapsed time: the effect requires roughly a decade of sustained exposure, not a short recent spell, and does not depend on treating that exposure as literally permanent and non-fading (§6.6). This does not mean Greek households are secretly optimistic; it means the "unexplained pessimism" reading of the residual does not survive once sustained exposure, not just its current level, enters the model. What remains genuinely open, beyond this paper's aggregate country-level data, is narrower: household composition, informal income, savings buffers, or institutional factors not captured in Eurostat's series may still account for a small remaining share.

6.11 Reporting heterogeneity: a robustness check against cultural-premium explanations

A distinct objection to §5–6 as a whole has not yet been addressed directly: subjective poverty is a self-report, and cross-country differences in how households answer such questions are a documented phenomenon in the survey-methods literature, independent of material conditions (Angelini et al., 2014, using anchoring vignettes to show that raw cross-country rankings of life satisfaction shift once individual-specific response-scale biases are corrected for). If part of Greece's outlier status reflects a stable, country-specific reporting style — rather than the material mechanisms identified in §6 — then this paper's central results would need to be reframed as measurement artifact rather than hardship. This subsection tests that possibility directly with six checks, framed as reporting *heterogeneity* rather than a claim about "Greek pessimism" as a trait, consistent with the individual-differences framing in the response-style literature itself. As a baseline sanity check, this paper's central subjective-poverty/AROP gap is not an artifact of this paper's own pipeline: Eurostat's own most recently published headline statistics independently show Greece's subjective poverty at 66.8% against an AROP rate of 19.6%, a 47.2-point gap and the largest of any EU member state (Eurostat, 2025) — consistent with, and derived independently of, the panel this paper's own results are built on.

Pre-crisis baseline, coverage-audited. Averaged over 2003–2008, Greece's subjective poverty was already among the EU's highest, an unbalanced-panel result that could in principle reflect EU-SILC's uneven early rollout rather than a genuine ranking. EU-SILC's own metadata documents that only six member states (Belgium, Denmark, Greece, Ireland, Luxembourg, Austria) held full coverage from the 2003 launch under a "gentlemen's agreement," with disruption in the series through 2005 as further states joined (Eurostat, 2026). Restricting the comparison to a properly balanced panel — the full-coverage six, a 2005–2008 25-country panel, and a 2007–2008 26-country panel — does not overturn the original ranking; it strengthens it. Among the six countries EU-SILC covered from inception, Greece ranks 1st, not 2nd as the unbalanced comparison implies. A pre-existing, stable component of the gap cannot be excluded on this evidence.

Difference-in-differences on the widening. A stable reporting premium predicts a stable gap over time; Greece's subjective-minus-AROP gap instead widened from

30.4 points (2003–2008) to 50.2 points (2019–2024), the largest widening of any EU country against a median *narrowing* of −8.8 points elsewhere. A two-way fixed-effects specification (country and year fixed effects, a Greece×post-2009 interaction, restricted to the 2007–2025 panel identified as near-fully-balanced above) returns a coefficient of +20.3 ($p<0.0001$). Given the single-treated-unit design, cluster-robust inference alone is not treated as sufficient here: an event-study decomposition shows no pre-existing differential trend (2007 coefficient −0.96, $p=0.17$, rising to significance only from 2011 onward); the estimate is stable to the treatment date (2009/2010/2011: 20.3/21.0/22.3) and to dropping any single control country (range 19.9–20.6); a periphery-only comparison group (Italy, Spain, Portugal, Cyprus, Malta) returns a larger coefficient (+24.5, $p<0.0001$); and a country-by-country placebo/permutation test — treating each of the 27 EU countries in turn as if it were "Greece" — ranks Greece's own coefficient 1st of 27, for an empirical p-value of 0.037. This is a country-placebo/permutation reference distribution, and its interpretation depends on treating the other 26 countries as exchangeable placebo units; Greece's crisis was not randomly assigned, so this is not randomization inference in the design-based sense. The rank and the 1/27 calculation are exact; the inferential reading is the part that requires the exchangeability assumption. It is reported in preference to asymptotic cluster-robust p-values, which are unreliable with a single treated cluster (Cameron & Miller, 2015, for the general concern; this paper's own permutation procedure is a direct, non-parametric response to it). A synthetic-control check was attempted and is *not* reported as corroboration, because it does not support the weight that would place on it. On the only substantively defensible pre-period, 2003–2008, it fails: just six countries have complete coverage in those years and the fit is poor (pre-period RMSE 25.3, with synthetic Greece at 2.7–8.1 against an actual 27.3–34.4), so the post-period divergence sits on top of a gap that already existed before the crisis. Restricting the match to the 2007–2008 window does produce a near-exact pre-period fit, but that window supplies two observations against roughly twenty-five free donor weights, where an exact fit is close to mechanically guaranteed and carries no evidential content. The comparative-case question is therefore treated as open rather than answered; a properly specified design with a justified pre-period, predefined donor eligibility, matching on multiple pre-period outcomes and objective predictors rather than only two outcome-gap observations, and RMSPE-ratio placebo inference is set out as future work in `docs/project_description_v3.md`. Nothing in the paper's conclusions rests on it.

Domain-specificity across self-report indicators. A *generic* response-style account predicts uniform extremity across all self-reported wellbeing measures; the data instead show a measure that tracks specifically with financial self-assessment. Greece ranks 1st of 27 on subjective poverty and 1st of 27 on financial expectations in every year with EU-wide coverage, but only 2nd–6th of 27 on general life satisfaction, never the most extreme. Standardizing all three to a common z-score scale (sign-harmonized so higher always indicates greater deviation from the EU mean) confirms this is a magnitude effect, not only a ranking one: Greece's average deviation is 3.76 on subjective poverty and 3.20 on financial expectations, versus 1.17 on life satisfaction — roughly a threefold difference. A country-specific response-style effect operating uniformly across self-report domains would not be expected to produce this pattern. The inference is bounded accordingly: domain specificity excludes a *generic* response-style account, and does not exclude a *financial-domain-specific reporting difference*. Such a difference — arising, for example, from perceived unfairness in taxation or public spending — would also be concentrated in financial items and absent from life satisfaction, and this evidence cannot distinguish it from materially grounded hardship. That competing account is set out in `docs/project_description_v3.md` §12a and remains open.

Two further, Greece-specific literature results corroborate rather than undercut this section's reading. Katsikas et al. (2015), covering 1995–2013 for the Crisis Observatory/ELIAMEP, and Zambarloukou (2015), characterizing Greece's welfare state as a structurally distinct South European model both before and after the crisis, independently document (i) a comparatively thin, family-reliant social safety net predating the crisis, consistent with a genuine structural baseline rather than a reporting artifact, and (ii) material poverty deterioration after 2009, particularly once measured against a fixed rather than moving threshold — independent, non-Eurostat corroboration of the direction, though not the exact magnitude, of §5–6's own findings.

ROBUSTNESS VERDICT

Greece entered the crisis with an unusually high reported level of financial difficulty, so a persistent country-specific reporting component cannot be excluded. But a stable reporting tendency cannot explain the subsequent deterioration and persistence by itself: the crisis-era widening is the largest of any EU country and survives the difference-in-differences robustness checks, while the attempted synthetic-control design failed and contributes no supporting evidence; the extremity is concentrated in financial self-report measures specifically rather than distributed across all self-reported wellbeing, and independent Greece-specific literature corroborates both the structural baseline and the post-2009 material deterioration. This subsection is retained as a robustness/context check on §5–6's results, not as an alternative explanation displacing them; it would be elevated to a primary finding only if the pre-crisis baseline component proved substantial and stable under further scrutiny, which the evidence gathered here does not indicate.

Full derivation, exact test statistics, the balanced-panel coverage audit, and the underlying literature review are reproducible from `scripts/40_reporting_style_robustness.py`, `scripts/41_reporting_style_robustness_v2.py`, and `scripts/42_reporting_style_robustness_v3.py`, logged in full in the companion technical report's `publication_strategy.md`.

7. Discussion

Sections 5 and 6 establish two things: that a substantial share of Greece's subjective-poverty exceptionalism is a measurement artifact of a relative-income threshold that resets every year, and that among the material conditions this paper can observe, *accumulated exposure* — not current-year conditions, and not incidence over duration alone — is what most fully separates Greece from the rest of the EU. This section first draws a general methodological lesson from that pattern (§7.1), then interprets the measurement and outlier results together (§7.2), then adds one further, more speculative dimension: what the documented record of Greece's financial-assistance programs suggests about why a recovery of this specific shape occurred (§7.3). That policy-history material is offered as context, not as an additional tested result, and is scoped narrower and hedged more heavily than §5–6 throughout.

7.1 A methodological caution for crisis contexts

The moving-threshold mechanism behind §5.2 is well established in the poverty-measurement literature (§3.1); this paper's contribution is a concrete illustration of its scale. During Greece's crisis decade, AROP moved far more slowly than living standards did, because the benchmark it is measured against was collapsing at the same time as the thing it measures — a structural property of any relative poverty line during a prolonged, broad-based macroeconomic collapse, not a flaw specific to Greece or to this paper's method. The practical implication generalizes beyond this case: in a country undergoing sustained macroeconomic contraction, AROP reported alone can understate deterioration for years at a time. This paper's own residual makes the same point from a different angle — even AROPE and a rich set of hardship controls leave real ground unexplained until a cumulative, exposure-based measure is added (§6.6). The recommendation that follows is not to discard AROP, which remains a useful, standardized, internationally comparable measure, but that in crisis or post-crisis contexts specifically, it is safest interpreted alongside a fixed-standard measure (an anchored poverty line, §5.3), a broader official measure (AROPE), and some account of accumulated exposure over time, rather than read on its own. Because this recommendation follows from a structural property of relative poverty measurement rather than from any Greece-specific substantive finding, it is expected to generalize more readily to other prolonged-crisis cases than any individual result in §6.

7.2 What the measurement and outlier results suggest together

Taken together, §5 and §6 point away from a reading of Greece's subjective poverty as households reacting to each year's economic news, and toward a population whose material conditions have been substantially, measurably degraded relative to the rest of the EU for a long time. The year-over-year sensitivity results (§6.7) reinforce this directly: Greek subjective poverty tracks its own AROP income-poverty rate unusually closely but barely moves with year-to-year swings in unemployment, housing costs, or material deprivation — consistent with a persistently elevated level rather than a reactive one. The relative-income poverty measure this paper's Part I results critique was, by construction, not built to register that kind of slow-moving, structural deterioration: a threshold that resets to 60% of a collapsing median tracks the crisis's average, not its duration or its accumulation. Long-term unemployment

(§6.5) is the sharpest single *current-level* expression of that duration in this paper's data; cumulative excess unemployment (§6.6) goes further, showing that it is the accumulated weight of that duration — not merely its most recent level — that most fully accounts for Greece's residual. This is one reason §7.3 asks whether the historical record offers a plausible institutional account of why sustained, rather than merely deep, hardship ended up carrying so much of the explanatory weight.

A further point worth making explicit: §6.9's salaried-only result shows the paradox is not reducible to an unemployment story that §6.5–6.6 simply re-describe in aggregate. Among people with jobs — a population by definition excluded from any unemployment-duration mechanism — official income poverty is unremarkable (17th of 27 on AROP) while subjective poverty remains the EU's highest by a wide margin. That the same pattern recurs among the employed, accompanied by a documented and independently significant imbalance between hours worked and hourly reward, indicates the residual this paper traces to accumulated labor-market exposure among the unemployed and the structural hours/reward imbalance among the employed are best read as two distinct, population-specific expressions of the same broader argument — that Greece's material conditions are degraded in ways a snapshot, single-year income measure does not register — rather than as a single unified mechanism. Consistent with §6.9's own within-country tests, this second expression is cross-sectional rather than dynamic: it says something stable about who is affected, not what changed for them from one year to the next, and the two should not be conflated when interpreting the paper's overall argument.

7.3 Policy history as one contextual dimension

Greece required three successive EU/IMF/ESM financial-assistance programs between May 2010 and August 2018 — each, in Pagoulatos's (2018) account, drawing "its existence from the failure of the previous."

Table 5. Greece's three financial-assistance programs. Source: Pagoulatos (2018), Table 2; European Stability Mechanism (2020).

PROGRAM	PERIOD	SIZE	LENDER(S)	PRIMARY FOCUS
1st (GLF/IMF)	May 2010–Mar 2012	€110bn (€72.8bn disbursed)	Bilateral loans + IMF	Front-loaded fiscal consolidation
2nd (EFSF/IMF)	Mar 2012–2014/15	€130bn+€34.5bn	EFSF + IMF	Private-sector debt restructuring (PSI); competitiveness reform
3rd (ESM)	Aug 2015–Aug 2018	up to €86bn (€61.9bn disbursed)	ESM	Milder fiscal path; institutional and financial-sector reform

Two independent sources — an outside academic assessment (Pagoulatos, 2018) and the lending institution's own commissioned, independent evaluation (European Stability Mechanism, 2020) — were deliberately consulted together for this section, and reach a convergent, two-sided verdict rather than a one-sided one. Both credit the programs with avoiding a disorderly Grexit, eventually meeting and then exceeding their fiscal targets, eliminating the current-account deficit, and delivering substantial structural reform (Greece's World Bank *Doing Business* rank improved from 96th of 181 to 61st of 189 between 2009 and 2015). Both are equally explicit about where the programs, in their own assessment, fell short, and attribute part of that shortfall to design choices rather than to the initial imbalances alone: fiscal consolidation was heavily front-loaded on a multiplier assumption the IMF's own subsequent research revised sharply upward (Blanchard & Leigh, 2013); debt restructuring was delayed by several years on liquidity-versus-solvency grounds Pagoulatos (2018) attributes to creditor bank exposure rather than to an assessment of Greek fundamentals; and labor-market liberalization proceeded well ahead of the safety net, with Pagoulatos (2018) reporting that under 20% of Greece's unemployed received any allowance during a period when roughly 70% had already been out of work a year or more — a gap the ESM evaluation independently describes as prioritizing "fiscal targets over growth-enhancing... reforms" at the level of program strategy.

Two of these documented design features line up, as a matter of timing and population, with this paper's own results, and are offered here as a plausible historical account worth naming — not as a tested mechanism. The front-loaded consolidation, applied through the years of deepest output loss, is consistent with the unusual depth of Greece's initial GDP collapse (26.2% from the 2008 peak to a first trough in 2013) and with the fact that the recovery which followed was still incomplete when a second, unrelated shock (the 2020 pandemic) arrived — leaving Greece sixteen years on, still not fully recovered, against a median EU recovery of three (§6.7). And the specific population left largely outside unemployment support — the long-term unemployed — is the same population whose size is this paper's single strongest predictor of Greece's subjective-poverty outlier status (§6.5). Neither connection is estimated quantitatively here; both would require data (macroeconomic counterfactuals, benefit-receipt microdata) outside this paper's aggregate-panel design (§4.4). Consistent with the two sources' own framing rather than a stronger claim on this paper's part: Pagoulatos (2018) titles his assessment "a qualified failure," and the ESM's own evaluators reach a similar bifurcated verdict from the lender's side — financial-stability objectives substantially met, growth and social objectives not. That an outside critique and the lender's own commissioned evaluation converge this closely is worth noting as one reason to treat the qualified reading as more than a partisan one, without this paper treating either source's account of *why* as established beyond what each source itself claims.

8. Limitations

- **Aggregate, not micro, data.** Every result in §5–6 is built from Eurostat's published country-year aggregates. Within-country heterogeneity, compositional effects, and informal income are invisible to this design (§4.4).
- **The anchored-poverty reconstruction is an approximation.** Built from four summary points rather than microdata, it is least reliable in exactly the years (2012–2014) where it shows the largest effect (§5.3).
- **Cross-country independence.** Country-clustered standard errors address some, but not all, of the non-independence risk Claessens et al. (2023) identify in cross-national panels of this kind; this paper has not tested its own residuals for spatial structure.
- **No causal identification.** The panel regressions establish conditional association, not causation. This applies with particular force to §7: the connections drawn there between program design and this paper's empirical findings are offered as a plausible, literature-grounded interpretation, not as a tested causal

chain, and no counterfactual (what would have happened under an alternative program design) is estimated.

- **Secondary sources for the policy history.** Section 7 relies on two synthesizing accounts (Pagoulatos, 2018; European Stability Mechanism, 2020) rather than primary program documents, troika review texts, or original interviews. Both sources are treated as credible on their own terms — one independently peer-circulated academic work, one an institutionally commissioned but editorially independent evaluation — but neither is a primary archival source.
- **Financial expectations and pessimism.** Any variable measuring households' own beliefs about their future finances is conceptually close to the subjective-poverty outcome itself; this paper treats such variables (§6, Model E; the institutional-trust context in §6.7) as the most heavily caveated evidence in the paper, not as independent economic conditions.
- **The work-effort-squeeze mechanism (§6.9) is cross-sectional, not dynamic, evidence.** It fails both a country-fixed-effects specification ($p=0.34$) and a first-differenced specification ($p=0.73$), and is uncorrelated with Greece's own within-country time series ($r=-0.03$). It is retained as FDR-corrected, genuine structural evidence that the paradox recurs among the employed specifically, not as a mechanism of comparable evidentiary weight to long-term or cumulative excess unemployment (§6.5–6.6), and is deliberately excluded from the §6.4 scorecard for this reason.
- **AROPE cannot be reconstructed from this paper's results.** AROPE is a union of three conditions whose household-level overlap Eurostat does not publish; §6 tests AROPE's underlying multidimensional-hardship premise one variable at a time, but no combination of those variables constitutes a reconstruction of AROPE itself (§5.1, §6 introduction). This is a structural limitation of working with aggregate, country-level data rather than EU-SILC microdata, not a gap this paper's design can close.
- **Cumulative-measure construction choices.** The cumulative excess-unemployment and duration measures in §6.6 depend on baseline-year and flooring choices, documented and tested for robustness (leave-one-country-out stability, a replacement test against long-term unemployment, rolling-window and decay-rate alternatives, and Benjamini-Hochberg correction across the full

18-candidate cumulative/duration family, computed from full-precision p-values inside the screening script itself) but not exhaustively varied against every alternative construction. The negative out-of-sample residual this measure produces is a property of the richest specification tested, not evidence that the paper's baseline findings overstate Greek hardship.

- **Candidate selection was not independent of Greece's own data.** Cumulative excess unemployment was chosen as the preferred candidate using the full panel, Greece included. Leave-one-country-out validation (§6.6) confirms the resulting coefficient is stable once this variable is fixed as the specification; it does not confirm that the choice of variable itself was unaffected by Greece's data point. A direct check — rerunning the 18-candidate screening with Greece dropped from the panel entirely — found the wage-duration measure edges ahead of cumulative excess unemployment ($p=0.0036$ vs. $p=0.0064$ without Greece in the screening data). Both remain the two strongest candidates either way, but the ranking between them is not robust to this choice. The stronger fix — repeating candidate selection separately within each of the 27 leave-one-country-out training folds, then scoring each fold's own selection against its held-out country — has now been performed (§6.6): the same variable is selected in 25 of 27 folds (both exceptions from the same duration/exposure family), and under full nested cross-validation Greece's residual is +2.70 points, 19th of 27 on prediction error. Selection stability and nested predictive validity are therefore both established; what no fold-based check can restore is pre-specification itself, which is why §4.3 classifies this as post-selection, robustness-validated evidence rather than a pre-planned confirmatory test.
- **The permanence of the accumulation is not established, only sustained exposure over roughly a decade.** Because the cumulative sum can only grow, it cannot distinguish genuine non-fading accumulation from a proxy for elapsed time since the crisis began. Rolling-window and decay-rate alternatives (§6.6) show a horizon of about ten years reproduces or exceeds the permanent sum's fit, while horizons of 3–5 years do not. This paper's claims are scoped to sustained exposure over approximately the past decade, not to literal permanent, non-fading accumulation since one specific baseline year.
- **Level of analysis for the cumulative-exposure mechanism.** The individual-level literature this paper draws on for the cumulative-exposure

mechanism (Lucas et al., 2004; Mousteri et al., 2018; Clark & Lepinteur, 2019) measures personal unemployment histories and individual outcomes; §6.6's own variable is a country-year aggregate. The aggregate association is consistent with the same underlying mechanism but is not direct evidence that the Greek households driving the subjective-poverty rate personally experienced sustained joblessness — an ecological-inference gap this paper's aggregate Eurostat data cannot close.

- **The age/household-composition finding (§6.8) is descriptive, not causal, and three sub-breakdowns were checked and found unavailable.** No age or household-composition variable enters any regression in this paper; the finding is a distributional decomposition of published Eurostat age-group series, not a tested mechanism. Sex within single-elderly households, an outcome-rate cross-tab of age or household type with tenure, and formal sampling-uncertainty figures were each checked directly against Eurostat's actual published dimensions and found not to exist as genuine statistics, rather than approximated by combining separate marginal tables. The decade-long trend is treated as the load-bearing claim; the single-year 2024-to-2025 movement, drawn from one EU-SILC survey wave, is corroborating rather than independently definitive.

9. Conclusion

Greece's exceptional subjective-poverty rate is neither a measurement artifact nor evidence of an unexplained national mood. AROP's moving threshold understated the income collapse; AROPE narrows but does not close the gap; housing and cash-flow strain narrow it further; and replacing headline unemployment with long-term unemployment cuts the out-of-sample residual from 11.6 to 3.9 points. In the fixed cumulative specification the residual reaches −0.8. Under full nested cross-validation — candidate selection and estimation repeated without the held-out country — Greece's residual is +2.70 points, 19th of 27, with eight countries predicted less accurately. The conservative conclusion is therefore not that one uniquely selected variable solves the paradox, but that duration and accumulated labor-market exposure remove Greece's exceptional outlier status. Financial-assistance-program design offers historically grounded context, not a tested causal chain. More broadly, AROP should be read alongside a fixed-standard threshold, AROPE or deprivation indicators, and duration-sensitive measures during prolonged macroeconomic collapses. Household composition, informal income, savings buffers, and institutional factors remain open; all results are associational and country-level.

10. Data, code, and AI-assistance disclosure

Data and code availability. Core data originate from the public Eurostat SDMX-JSON API; no core panel series is hand-entered. The dependency-aware Makefile runs the numbered extraction, cleaning, and modeling scripts. `make verify` reproduces all 41 published checks from the archived source-data snapshot. `make reproduce` exports an isolated tree, re-fetches all raw inputs, rebuilds the outputs, and compares them with that publication vintage. The 2026-08-20 live run completed every acquisition and analysis step, then correctly stopped on one revised-source difference: fresh real-wage data produce 18 rather than 17 detrended FDR survivors; the other 40 checks agree. This distinction is expected when an external statistical agency revises historical observations and is reported rather than hidden by silently updating the expected value. Institutional-trust figures and policy-history sources are public literature context rather than modeled pipeline inputs. Core vintage: 2026-08-19; work effort: 2026-08-20.

AI-assistance disclosure. This paper's underlying data retrieval, statistical modeling, and drafting were carried out with the assistance of Claude (Anthropic), an AI system, operating under direct human direction throughout: every analytical choice (model specification, variable inclusion, validation method), every claim's wording, and every citation was reviewed by [the author], who takes full responsibility for the content, interpretations, and conclusions of this paper. Two sources cited in §7 (Pagoulatos, 2018; European Stability Mechanism, 2020) were retrieved and read in full by the assisting AI system directly from their original PDF publications as part of this process. [Author: confirm, edit, or remove this disclosure before submission, per your target venue's policy.]

[Open items for the author before this draft is submission-ready: (1) author name and institutional affiliation, currently placeholders throughout; (2) confirm single combined-paper structure versus splitting Parts I and II into separate submissions; (3) confirm APA citation style, used as a default here; (4) select a target venue (working-paper series, preprint server, or journal) and reformat to its house style; (5) if targeting a specific journal, verify each cited paper's exact venue and page numbers directly against the publisher record before submission, since several citations here were carried forward from the companion technical report's own literature review rather than re-verified against publisher metadata for this draft.]

A. Cumulative-hardship validation details

This appendix preserves the full candidate-screening and robustness record behind §6.6 while keeping the main argument focused on the estimand and the conservative nested result.

A.1 Full cumulative-hardship construction and validation record

Long-term unemployment (§6.5) is a snapshot: the share of the labor force currently out of work for 12 months or more. It does not distinguish a country that had one brutal year and mostly recovered from one that has had many mildly-bad years and never recovered at all. This section tests a cumulative alternative directly: for each country and year from 2009 onward, that year's unemployment rate minus the country's own 2009 baseline rate, floored at zero (so a below-baseline year contributes nothing rather than offsetting a later bad year), summed cumulatively through 2024. The result is each country's running total of excess unemployment-years carried since the crisis began — a measure of accumulated exposure rather than current level.

This construction extends, rather than departs from, a dynamic-poverty paradigm the EU statistical system already recognizes at the individual level. Eurostat's own persistent-poverty indicator counts a person poor only if they are AROP-poor in the current year *and* in at least two of the preceding three years, precisely because a single-year snapshot is known to understate how widespread and durable poverty experience actually is; a 2026 European Commission study of EU-SILC longitudinal data across 2011–2022 concludes that "poverty experience is more widespread and heterogeneous than standard indicators suggest" and explicitly recommends dynamic measures for policy design (Directorate-General for Employment, Social Affairs and Inclusion, 2026). Cumulative excess unemployment applies the same logic one level up: rather than tracking whether a household was poor in multiple recent years, it tracks whether a country's labor market has stayed depressed relative to its own baseline for multiple years running. At the macroeconomic level specifically, this mirrors the hysteresis mechanism the IMF documents for the euro area — prolonged unemployment eroding skills and detaching workers from the labor force in ways that compound rather than reset year to year (Lin, 2016). To the best of this paper's literature search, no existing EU-focused study uses a country-level cumulative-excess-unemployment index of this specific construction to explain the subjective-poverty gap; the underlying mechanism, however, is well established at both the individual level (below) and the macro level (Lin, 2016).

Adding cumulative excess unemployment to Model C-LTU (Model G) lifts the fit from $R^2=0.914$ to 0.930 and moves Greece's out-of-sample residual from 3.9 to −0.8 points — the model now slightly *overpredicts* Greek hardship rather than underpredicting it. Once this is the variable being tested, its coefficient is stable and positive across all 27 leave-one-country-out refits (range 0.134–0.197, no sign flips), and remains significant in the refit excluding Greece itself ($\beta=0.164$, $p=0.0064$). This establishes that the coefficient is not an artifact of any single country's data once cumulative excess unemployment is the specification being estimated; it does not by itself establish that Greece's own data was uninvolved in the earlier step of selecting this variable over its alternatives — that question is addressed directly below. A year-by-year check of the final model's residual for Greece (fit excluding Greece) shows small, mostly negative-to-near-zero residuals for nine of the ten years 2015–2024 (−3.25 to +1.01), with one exception (2021, +4.40, plausibly a pandemic-specific disruption) — evidence against a single-year artifact driving the average.

Two robustness checks establish that this is a genuine additional layer rather than long-term unemployment relabeled or an artifact of overfitting. First, a replacement test: substituting cumulative excess unemployment *for* long-term unemployment (rather than adding it) performs worse ($R^2=0.911$, out-of-sample gap 4.3) than either the long-term-unemployment baseline ($R^2=0.914$, gap 3.9) or the combined specification ($R^2=0.930$, gap −0.8); variance inflation factors in the combined model remain modest (maximum 4.5). Second, an equivalent construction was tested for the AROP poverty threshold itself — a cumulative threshold shortfall, the most narratively anticipated candidate given §5.2's moving-threshold finding — and returned a null result ($p=0.291$), reported here for completeness rather than omitted. A related duration measure, consecutive years Greek real wages remained below their own 2008 level, is independently significant ($p=0.0045$) and passes the same leave-one-out stability check (coefficient range 0.31–0.54, always significant, including excluding Greece: $\beta=0.31$, $p=0.0036$); the equivalent GDP-based duration constructions (years below 2008, years below own peak, longest continuous shortfall streak, cumulative negative-growth years) are all non-significant ($p=0.13$–0.89). Combined with cumulative excess unemployment, the wage-duration measure remains statistically significant ($R^2=0.931$, gap −1.1), but within Greece's own time series the two variables are highly correlated ($r=0.95$, versus 0.54 panel-wide) — too entangled in Greece's specific case to treat credibly as two independent mechanisms.

A negative out-of-sample residual requires careful interpretation. It does not indicate that Greek households are secretly more optimistic than their conditions warrant, and it should not be read as the gap being "more than fully" explained — a framing this paper deliberately avoids. It is a property of this specific, richest specification tested (Model C-LTU plus cumulative excess unemployment), not a claim about the paper's baseline finding. The more precise reading is that once accumulated labor-market exposure, not merely its current level, enters the model, the interpretation of Greece's residual as an unexplained "pessimism premium" no longer survives.

Because several related cumulative and duration constructions were screened together before this section's preferred candidate was selected — the original eight-variable cumulative battery (GDP, wages, and unemployment excess, on two baselines each, plus the AROP-threshold shortfall) and the ten-variable duration/direction battery (years below baseline, years below own peak, longest continuous shortfall streak, cumulative negative-growth years, for both GDP and wages) — this family of 18 exploratory tests was checked against Benjamini-Hochberg false-discovery-rate correction, computed from full-precision p-values inside the same script that ran the screening (§4.2's discipline, applied consistently). Cumulative excess unemployment survives comfortably (raw $p=0.00013$, FDR-adjusted $p=0.0024$); the wage-duration measure survives, but narrowly (raw $p=0.0045$, FDR-adjusted $p=0.0408$, just under the conventional 0.05 threshold); none of the other 16 candidates in the family survives correction (adjusted $p \geq 0.25$ in every case). This identifies cumulative excess unemployment and the wage-duration measure as the two candidates worth taking seriously; it does not, on its own, establish which of the two should be considered primary — that ranking depends on a further check reported next.

Selection sensitivity to Greece's own data. Cumulative excess unemployment was identified as the preferred candidate using the full 27-country panel, Greece included — a legitimate way to conduct the screening, since Greece's data is a genuine part of the historical panel and excluding it a priori would itself be an unmotivated choice. It is nonetheless worth checking directly whether that choice is sensitive to Greece's presence in the screening step, since leave-one-country-out validation (above) tests coefficient stability for an already-selected variable, not the independence of the selection itself. Rerunning the complete 18-candidate screening with Greece dropped from the panel entirely — excluded from model-fitting, not merely held out for evaluation — the wage-duration measure edges ahead of cumulative excess unemployment ($p=0.0036$ vs. $p=0.0064$ without Greece in the screening data; both were significant with Greece included as well, at $p=0.00013$ and $p=0.0045$ respectively). The set of candidates worth taking seriously is unaffected by this check: the same two variables are the two that matter, with or without Greece in the screening panel. Which of the two ranks first is not independent of that choice. This paper reports cumulative excess unemployment as the headline construction because it has the more direct interpretation (a labor-market variable, not one further step removed via wages) and remains the stronger FDR survivor under the panel composition used throughout the rest of this paper's models, but treats it and the wage-duration measure as two closely related, mutually reinforcing indicators of the same underlying mechanism rather than as one uniquely and unambiguously identified variable with a clear runner-up. The stronger version of this check — repeating the complete 18-candidate screening independently inside each of the 27 leave-one-country-out folds, with each fold's country dropped from screening entirely — has also been performed (`scripts/44_nested_selection_validation.py`). Cumulative excess unemployment is selected first in 25 of 27 folds; the Greece-excluded fold selects the wage-duration measure (the sensitivity documented above), and the Finland-excluded fold selects a closely related GDP-duration measure (with cumulative excess unemployment still ranked 3rd, $p=0.00045$); cumulative excess unemployment remains statistically significant in every fold (worst-case $p=0.0064$). Selection is therefore stable across folds, with both exceptions landing on near-neighbor duration/exposure constructions inside the same family rather than on unrelated candidates. In 26 of the 27 folds the selected winner also survives Benjamini-Hochberg correction applied within that fold across all 18 candidates.

Nested predictive performance. Selection stability is a weaker claim than predictive validity, so the procedure is also scored end to end: in each fold, that fold's *own* selected candidate is fitted on the 26 training countries and used to predict the held-out 27th, which took part in neither selection nor estimation. This is the genuine nested cross-validation quantity. Across the 27 folds the mean absolute prediction error is 4.53 points (median 3.82). For the Greece fold specifically — where the procedure selects the wage-duration measure rather than cumulative excess unemployment — the nested residual is **+2.70 points** (mean absolute error 4.51), placing Greece 19th of 27 on prediction error: **eight EU countries are predicted less accurately than Greece** by their own fold's selected model. This figure, not the −0.8 residual of the fixed Model G specification, is the appropriate one to cite when the claim concerns the whole select-then-fit procedure rather than one already-chosen variable. The two differ for a reason worth stating plainly: −0.8 comes from a specification whose variable was selected with Greece in the screening panel, while +2.70 additionally prices in the selection step. Both support the same substantive conclusion — that Greece ceases to be a cross-country outlier once accumulated labour-market exposure enters the model — but +2.70 is the more conservative and more defensible number, and the mid-pack ranking is the stronger form of the claim, since it no longer depends on which member of the duration/exposure family a given fold happens to pick. Full per-fold results: `nested_selection_validation_folds.csv`, `nested_selection_validation_detail.csv`; the single-exclusion comparison remains in `cumulative_hardship_selection_excl_greece.csv`.

Testing the permanence assumption. Because the cumulative sum is floored and strictly non-decreasing, it cannot on its own distinguish genuine undiminished accumulated exposure from simply encoding elapsed time since the crisis began — a trend proxy would show the same monotonic pattern. Five alternative constructions were tested from the same underlying per-year excess-unemployment series, each capable of falling over time: trailing 3-, 5-, and 10-year rolling sums, and exponentially-decayed running sums at two decay rates (20%/year, 10%/year). A 3-year window is not statistically significant ($p=0.29$); a 5-year window is borderline ($p=0.042$). A 10-year rolling window, however, fits at least as well as the permanent since-2009 sum on both criteria ($R^2=0.932$ vs. 0.930; Greece's out-of-sample gap −0.47 vs. −0.82; $p=0.00002$ vs. $p=0.00013$), and both decayed versions also fit strongly ($p=0.0003$ and $p=0.00006$ respectively), though with somewhat larger out-of-sample overshoot (−1.78 and −1.90). The honest reading is that a window as short as 3–5 years cannot reproduce the effect, ruling out a short recent bad patch as the explanation, but a horizon of roughly a decade reproduces or improves on the permanent sum's fit. This paper accordingly frames the mechanism as *sustained* excess unemployment over approximately the past decade, not literal permanent, non-fading accumulation since one specific baseline year — a claim strengthened, not weakened, by resting on convergence across several related constructions rather than on the single, somewhat arbitrary choice of an unbounded running sum. Full results: `cumulative_hardship_rolling_decay_battery.csv`.

The mechanism this section documents — that accumulated, not merely current, exposure to hardship depresses subjective well-being — has substantial support in the individual-level literature, though at a different level of analysis than tested here. Lucas, Clark, Georgellis, and Diener (2004), following more than 24,000 individuals in a 15-year German panel, find that life satisfaction does not fully return to its pre-unemployment baseline even after re-employment. Mousteri, Daly, and Delaney (2018), across 14 European countries, find that each additional six-month spell of past unemployment predicts lower life satisfaction after age 50, controlling for current conditions. Clark and Lepinteur (2019), using British cohort data, find that cumulative unemployment experience before age 30 predicts lower life satisfaction at 30, conditional on current unemployment status. All three studies measure individual employment histories and individual-level outcomes; this paper's cumulative excess-unemployment variable is a country-year aggregate, and its association with Greece's *aggregate* subjective-poverty rate is not direct evidence that the same Greek households experienced sustained personal joblessness across the period studied. The aggregate result is consistent with, and plausibly reflects, the same underlying scarring mechanism documented at the individual level — but this paper does not observe individual employment histories, and the two levels of evidence should not be conflated.

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This working paper is a preprint, not peer reviewed. It draws its empirical core (§5–6) from a companion technical report covering the same Eurostat-derived analysis in fuller, plain-language form, and its Discussion (§7) from two additional primary sources read in full for this draft (Pagoulatos, 2018; European Stability Mechanism, 2020). All Eurostat figures reflect an API pull dated 2026-08-19 (§6.9 work-effort series: 2026-08-20); see §10 for data-vintage and reproducibility notes.